

LAB NO : 03

LAB TITLE : Relational Operators, Logical Operators and Conditional Statements in C++.

OBJECTIVE :

The objective of this lab is to understand how relational operators, logical operators, and conditional statements work in C++. In this lab we learn how to compare values, evaluate logical conditions, and control program flow using if, if-else, and else-if statements.

THEORY :

In C++ programs normally execute instructions one after another. However, sometimes we need to make decisions depending on certain conditions. For this purpose C++ provides relational operators, logical operators, and conditional statements.

Relational operators are used to compare two values:

1. < Less than
2. > Greater than
3. <= Less than or equal to
4. >= Greater than or equal to
5. == Equal to
6. != Not equal to

Operator Meaning :

1. && Logical AND
2. || Logical OR
3. ! Logical NOT

Conditional statements allow a program to execute different blocks of code depending on conditions.

Types of conditional statements :

1. If statement
2. if-else statement

3. else-if statement

SOFTWARE :

1. DEV-C++ IDE
2. C++ Compiler

PROCEDURE / PROCESS :

Comparison Using Relational Operators :

1. Start the program
2. Declare two integer variables
3. Take two numbers from the user
4. Compare the numbers using relational operators
5. Display the comparison result
6. End the program

```
#include <iostream> // input aur output ke liye library
using namespace std; // standard namespace use karne ke liye

int main()
{
    int a, b; // do integer variables declare kiye gaye hain

    cout << "Enter two numbers: "; // user se do numbers input lene ka
message
    cin >> a >> b; // values input lena

    if(a == b) // comparison start
        cout << "Both numbers are equal"; // agar dono barabar hain

    else if(a > b)
        cout << "First number is greater"; // agar pe
```

Logical Operators :

1. Start the program Logical Operators
2. Declare an integer variable
3. Take a number from the user
4. Check if the number is greater than 0 and less than 100
5. Display the result
6. End the program.

```
#include <iostream> // input aur output ke liye library
using namespace std; // standard namespace use karne ke liye

int main()
{
    int num; // integer variable jisme user ka number store hoga

    cout << "Enter a number: "; // user se number input lene ka message
    cin >> num; // user se number lena

    if(num > 0 && num < 100) // check karna ke number 0 aur 100 ke darmiyan
hai ya nahi
        cout << "Number is between 0 and 100"; // agar number range ke andar
hai
    else
        cout << "Number is not between 0 and 100"; // agar number range ke
bahar hai

    return 0; // program yahan end hota hai
}
```

If Statement (Positive Number) :

1. Start the program.
2. Declare an integer variable.
3. Take a number from the user.
4. Check if the number is positive.
5. Display the message.

6. End the program.

```
#include <iostream> // input aur output ke liye library
using namespace std; // standard namespace use karne ke liye

int main()
{
    int num; // integer variable jisme number store hoga

    cout << "Enter a number: "; // user se number input lene ka message
    cin >> num; // user se number lena

    if(num > 0) // check karna ke number positive hai ya nahi
        cout << "The number is positive"; // agar number zero se bara hai

    return 0; // program end
}
```

CONCLUSION :

In this lab we learned how to use relational and logical operators in C++. We also learned how conditional statements such as if, if-else, and else-if help a program make decisions based on different conditions. These concepts are very important for writing logical programs..